Software Engineering II
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Final Project Definition and Delivery

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Congratulations on reaching the final stage of your *Software Engineering II* course project! This document outlines the requirements for your final project delivery. Your submission should demonstrate a complete, well-documented, and functional web application, integrating the concepts and practices learned throughout the course.

Final Project Scope and Objectives:

- **Web Application:** Develop a simple web application consisting of a REST API backend and a basic frontend (any technology).
- **Object-Oriented Programming (OOP):** Apply OOP principles in backend codebases.
- **Design Patterns:** Implement at least two design patterns (e.g., Singleton, Factory, Observer) in your project and document their usage.
- **Database Integration:** Use a relational database (e.g., SQLite, PostgreSQL, MySQL) to store and manage application data.
- **Software Testing:** Provide minimal automated tests for your backend (unit and/or integration tests).
- **CI/CD Pipeline (optional):** Set up a basic CI/CD workflow using GitHub Actions/GitLab, including steps for running tests and building a Docker image for your backend.

- **Documentation:** Deliver clear documentation covering architecture, design decisions, API endpoints, database schema, setup instructions, and CI/CD configuration (optional).
- **Reflection and Evaluation:** Critically evaluate your solution, discussing strengths, limitations, and possible future improvements.

Methodology and Deliverables:

1. **Documentation:**
   - Compile all sections (architecture, design patterns, API documentation, database schema, testing strategy, setup, deployment and evaluation) into a single, well-organized PDF.
   - Include diagrams, code snippets, and references as needed.
   - Implemented User Stories with Supporting Evidence.
2. **Project Repository:**
   - Organize all source files, scripts, diagrams, and documentation in a folder named `Final-Project` in your course repository.
   - Provide a `README.md` that explains the structure, setup, and usage of your project.
3. **Project Implementation:**
   - Deliver a functional implementation of your web application using the selected technology stack.
   - Ensure the implementation covers the main features, REST API endpoints, database operations, and frontend interactions described in your documentation.
   - Include clear instructions for setup, execution, and testing in your `README.md`.
4. **CI/CD Demonstration (optional):**
   - Provide a working GitHub Actions workflow file (`.github/workflows/ci.yml`) that runs tests and builds a Docker image for your backend.
   - Document the CI/CD process and show evidence of successful runs (e.g., screenshots, logs).
5. **Demonstration (Mandatory):**
   - Prepare and deliver a brief presentation or video (15 minutes) demonstrating your system's main features, API endpoints, frontend.
   - The demo is a required part of the final evaluation.

Project Requirements Checklist:

- API - backend (any technology).
- Basic frontend (any technology).

- OOP principles applied in code.
- At least two design patterns implemented and documented.
- Database integration.
- Minimal automated tests for backend.
- CI/CD pipeline with GitHub Actions and Docker image build (optional).
- Clear documentation and setup instructions.

Examples of Application Ideas:

- Task manager or to-do list application.
- Simple blog or content management system.
- Basic inventory or product catalog.
- Contact management or address book.
- Event registration or booking system.

Deadline:

- Second week of July: technical presentations.
- Third week of July: final functional presentations.

Notes:

- All documents must be in **English**.
- Cite any references (articles, tutorials, frameworks) that influenced your design choices.
- Focus on clarity, completeness, and professional presentation.
- This is your opportunity to showcase your ability to design, implement, test, and deploy a robust web application using modern software engineering practices.

Good luck! Your final project is the culmination of your learning and effort throughout the course. Make it count!